

When a Consciousness Verdict Cannot Be Earned: A Descent-Theoretic Admissibility Criterion

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Abstract

A claim that a system is or is not conscious is, in the end, a claim about evidence, only as good as the evidence’s power to tell consciousness apart from its absence. This is an identification problem. A verdict is licensed only when the target property can be recovered from a declared observation site, and where it cannot, more data does not help. For a structurally specified class of hidden-variable systems this recovery fails *necessarily*, surviving every admissible revision of the measurement language, not only the instrument in hand. The failure takes a single form, *revision-stable non-effective descent*: the target cannot be assembled from any admissible family of local measurements. The underlying factorization is proved for the finite deterministic and exact finite stochastic cases, with both factorization regimes, the finite kernel-pair lemma, and conservative invariance mechanically verified in Lean. The conservative-invariance result shows that no Morita-equivalent re-presentation of the measurement language removes the obstruction. The general reduction, that every admissible identification procedure factors through effective descent, is isolated as an explicit open obligation, not assumed. The criterion’s normative force is confined to fail-closed audit settings: where it holds, a low-concern verdict is not licensed, and the default is restriction. Non-identification is not thereby inadmissible for science in general. Prior category-theoretic work characterizes the structure of consciousness constructively. This paper runs the other way, giving a limitative criterion for when an attribution can be made at all. The criterion is worked through a minimal broadcast-report architecture, where its global-identity target lands in the $Z = 1$ class while access stays identified, and through an open-weight transformer. Explicit site constructions for full-scale Integrated Information, Global Workspace, and Higher-Order theories remain open.

Keywords: consciousness; scientific admissibility; identification; descent theory; artificial intelligence systems; philosophy of science

Technical summary

A hidden-variable system H generates revision-stable necessary non-identification of a target property Θ , under the descent-factorization obligation (Theorem 1, proved here for the finite cases, open in general), when the target fiber $F \rightarrow H$ has non-effective descent over the admissible observation site C_Y for every admissible structural revision $r \in \text{Rev}_\Omega(Y)$. The revision envelope $\text{Rev}_\Omega(Y)$ covers future structural changes to the observation framework, not only refinements within a fixed language. Theorem 1 recovers the kernel-pair obstruction lemma (Lemma 1) as the classical special case. Integrated Information Theory [Tononi et al., 2016], Global Workspace Theory [Baars, 1988, Dehaene and Changeux, 2011], and Higher-Order Theories [Rosenthal, 2005] are discussed as illustrative sketches of three positions relative to the criterion, with the explicit site and target fiber for each left open.

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1 Identification failure as a structural constraint

Suppose an AI company claims its model deserves a “low concern” verdict on moral status, meaning there is no significant probability the system is conscious. What evidence would settle this? The company offers behavioral evidence (the system does not exhibit distress signals), functional evidence (it lacks the neural correlates of consciousness), and architectural evidence (its computation is unlike human cognition). A regulator reads the evidence and asks whether it separates consciousness from an observational duplicate.

None of this connects the target to the evidence in the right way. A system that lacks consciousness can still duplicate behavioral competence. Neural-correlate evidence is tied to human neural architecture. Architectural dissimilarity shows that two systems differ, while leaving open whether either of them lacks consciousness. So the proposed tests leave two stories standing, and no experiment on the table separates them.

Philosophers call this the “hard problem.” Stated more precisely, it is an *identification problem*. A parameter is identified when different true values yield different observable distributions. Once identification fails, more data does not help.

The central question is whether identification failure for consciousness holds *necessarily*, surviving admissible improvements in measurement, or only *contingently*, so that a better admissible test could remove it. The answer decides whether a failed test should prompt further measurement or a restricted verdict.

For a well-defined class of systems the failure is necessary relative to the declared observation site and revision envelope, conditional on the descent-factorization obligation stated in Theorem 1 and proved below for the finite deterministic and exact finite stochastic cases. A system in this class cannot receive an auditable low-concern verdict without a successful identification certificate.

2 Formal preliminaries

“Admissible observations” names the measurements permitted for the system under evaluation. Two hidden states are experimentally indistinguishable when they agree under all of those observations. And the *experimentability witness* records the minimal typed specification a scientific consciousness claim must provide. A theory that cannot supply this witness has made no falsifiable claim, so the theorems below leave it untouched.

For the setup, let H be the set of possible hidden states of a system (its internal state space: neurons firing, weights activating, whatever physical process underlies behavior). Let $\Theta : H \rightarrow T$ be the target map (e.g., “is this state conscious or not?”). Let Y be the collection of admissible observations: every measurement, perturbation, imaging technique, and behavioral test permitted. Let $E_Y : H \rightarrow \mathcal{R}_Y$ be the observation function mapping hidden states to observable records.

Definition 1 (When two hidden states look identical). *Hidden states h_1 and h_2 are experimentally indistinguishable if every admissible measurement gives the same distribution:*

$$h_1 \equiv_Y h_2 \iff \sup_{e \in \text{Eff}(Y)} |p(e \mid h_1) - p(e \mid h_2)| = 0.$$

On classical deterministic systems this simplifies to $E_Y(h_1) = E_Y(h_2)$.

Definition 2 (Experimentability witness). *A theory (H, Y, E_Y, Θ) is experimentable if it supplies a typed specification $W = (\mathcal{C}, \mathcal{R}, \mathcal{T}, H, Y, E_Y, \Theta)$ in which $\mathcal{R}$ and $\mathcal{T}$ have decidable equality. By*

“decidable equality” we mean a procedure that definitively answers “did these two records match?” and “did these two target values match?” Absent such a specification, the theory makes no falsifiable scientific claim in the sense used here.

Remark 1. Faced with a theory that cannot supply the witness, a governance regulator should require the witness before deliberating about the theory at all. IIT, GWT, and HOT occupy different positions under this test, as Section 9 shows: IIT does not yet supply the witness for full Φ -based identification, whereas GWT and HOT can supply witnesses for narrower targets.

3 Structural characterization of the non-identification class

The hidden-variable systems that fall in the necessary non-identification class are picked out by four structural properties, none of them mentioning consciousness. A system satisfying all four generates $Z = 1$ by the theorem in §5, whose machinery §§4–5 develop.

Definition 3 (P1, local contextual presentability). *A family of admissible local contexts $\{U_\alpha \rightarrow H\}$ exists such that each U_α supports local target assignments. The target Θ is partially evaluable at the level of individual contexts, even when not globally evaluable.*

Definition 4 (P2, nontrivial transition structure). *On overlapping contexts $U_\alpha \cap U_\beta$, local target assignments are related by transition maps $g_{\alpha\beta} : F_\beta|_{U_{\alpha\beta}} \rightarrow F_\alpha|_{U_{\alpha\beta}}$ that satisfy local compatibility on every pair but fail global compatibility across all contexts simultaneously.*

Definition 5 (P3, nontrivial holonomy). *The context groupoid $\Pi_Y(H)$ acts on the target fiber F with no admissible global fixed section: $\text{Fix}(\rho) = \emptyset$. The descent datum is non-effective: no global section is compatible with all local data simultaneously.*

Definition 6 (P4, equivariant admissible operations). *Every admissible intervention or observation refinement $Y' \in \text{Adm}(Y)$ preserves the holonomy structure: $\rho_{Y'} \circ a = a \circ \rho_Y$ for every admissible action a . No expansion or refinement of the observation family dissolves the obstruction.*

Properties P1–P3 fix a static obstruction at one observation level. P4 makes the obstruction persist, since it guarantees that the holonomy survives every admissible intervention. Where P4 fails, the obstruction is contingent, and a better instrument can in principle remove it. Where all four hold, the obstruction is necessary, beyond the reach of any improvement in measurement methodology.

These properties depend only on the topology of the causal graph and on how the target fiber descends over the observation site. Substrate, material composition, and biological origin drop out. Any physical system, biological or digital, that instantiates equivalent P1–P4 structure falls in the same identification class.

P1–P4 need not exhaust every possible source of the hard problem. For a system with P1–P4, conservative expansions of the structural language draw no new structural distinctions, so they cannot supply the sufficiency bridge from non-identification to consciousness attribution, and the obstruction is stable under conservative re-presentation. Non-conservative expansions can matter, yet they change the admissible site Y and must be declared in the revision envelope Ω . The boundary result below concerns this conservative-stability claim, and a complete taxonomy of possible consciousness obstacles lies outside it.

4 The kernel-pair non-factorization lemma

Identification can fail in a way that is at once the simplest and the easiest to miss. Two hidden states look identical from outside, since every measurement returns the same result, yet they differ

in whether Θ holds. In the consciousness case, a conscious system and a non-conscious system produce identical observable records under every admissible test, so no experiment distinguishes them. This is the kernel-pair obstruction.

Definition 7 (Kernel pair). *The kernel pair of E_Y is the set of all experimentally indistinguishable pairs:*

$$K_Y = \{(h_1, h_2) \in H^2 : h_1 \equiv_Y h_2\}.$$

Think of K_Y as the observation window's blind spot. The quotient $H/\sim_Y$ is the compressed state space in which indistinguishable states are treated as one.

Lemma 1 (Kernel-pair obstruction). *For an experimentable theory (H, Y, E_Y, Θ) :*

Θ is necessarily non-identified if and only if there exist hidden states h_1, h_2 that look identical from outside ($h_1 \equiv_Y h_2$) but differ in the target ($\Theta(h_1) \neq \Theta(h_2)$).

Equivalently: Θ does not factor through the experimentally compressed state space $H/\sim_Y$.

Proof sketch. If the pair exists, no estimator on observable records can separate h_1 from h_2 . If no such pair exists, Θ is constant on each indistinguishability class and is identified. $\square$

This is the standard non-identification condition of [Pearl \[2009\]](#), instantiated in the present typed setup.

A concrete check follows. Build a six-state system $\{a, b, c, d, e, f\}$. Let the observation function collapse $\{a, b\}$ to the same record. Set $\Theta(a) = 0$, $\Theta(b) = 1$. The kernel pair (a, b) fires: identical observation, different target, with residual ambiguity $\log_2 2 = 1.000$ bit, and no symmetry or automorphism enters. The argument turns on plain indistinguishability, with nothing gauge-theoretic about it.

The lemma falls short in one respect. Lemma 1 needs the compressed state space $H/\sim_Y$ to be a well-behaved mathematical object. On quantum systems, systems with non-abelian symmetries, or infinite-dimensional spaces, the quotient may fail to be constructible. Theorem 1 handles all of these.

5 The descent-obstruction theorem

Lemma 1 catches the case where two states look identical and the target differs. A more general obstruction lurks beyond it. Every *pair* of measurement contexts can be locally consistent while no single global assignment fits all contexts at once. Quantum mechanics knows this phenomenon as contextuality, and it carries over to consciousness attribution. Theorem 1 states the local-to-global obstruction at the level of effective descent.

5.1 Limits of Čech cohomology

[Abramsky and Brandenburger \[2011\]](#) developed the standard tool for contextuality, sheaf cohomology, for quantum systems. A nonzero cohomology class $\check{H}^1(\mathcal{U}_Y, \hat{\Theta}) \neq 0$ certifies that local target assignments cannot be glued globally, detecting cases that Lemma 1 misses.

Yet $\check{H}^1$ does not tell the whole story. It demands a good cover (geometric regularity), abelian target structure (with addition and subtraction defined), and an obstruction lodged in exactly degree 1.

Quantum POVM systems, non-abelian symmetry groups, and systems with non-standard topologies all violate one or another of these conditions.

The kernel-pair case (Theorem 1) is a direct counterexample, with $\check{H}^1 = 0$ but $D = 1$. A detector that checks only $\check{H}^1$ recovers half the cases on a minimal two-case suite, recall = 1/2, which is too coarse to rely on.

5.2 Effective descent

Imagine reconstructing the value of Θ by combining local measurements from overlapping contexts. *Effective descent* means that some procedure takes all the local results, checks them for overlap consistency, and assembles them into a single global answer. When descent fails to be effective, no such assembly exists: every pair of local measurements can agree, and the global assembly still goes through nowhere.

Definition 8 (Descent-obstruction flag). *For admissible observation family Y' :*

$$D_{Y'}(X) = \begin{cases} 1 & \text{if } F|_{\mathcal{C}_{Y'}} \text{ has non-effective descent} \\ 0 & \text{otherwise.} \end{cases}$$

Definition 9 (Revision envelope). *Let Ω denote the declared physical, computational, and operational constraints on what structural changes to the observation/intervention framework are admissible. A structural revision r is any change to the admissible site that falls within Ω . $\text{Rev}_\Omega(Y)$ denotes the set of such revisions. A revision is conservative (Morita-equivalent) if it induces an equivalence of structural model categories over the same admissible site, preserving the target fiber, covering families, and context structure. A revision is non-conservative if it adds a genuinely new admissible observable, intervention, or target-relevant operation not definable from the old site.*

Definition 10 (Revision-stable persistent non-effective descent). *The identification obstruction is revision-stable if it survives every admissible structural revision:*

$$Z = f(X, Y) = \min_{r \in \text{Rev}_\Omega(Y)} D(r_*X) = 1.$$

When $Z = 1$, no measurement protocol, refinement, expansion, or structural revision of the observation family within Ω lets the target be assembled globally from local data. The earlier formula $\min_{Y' \in \text{Adm}(Y)} D_{Y'}(X)$ is the special case in which Ω holds only refinements within a fixed structural language. Definition 10 reaches all admissible structural revisions.

Proposition (Conservative Invariance). *If $e : \mathcal{L}_D \rightarrow \mathcal{L}_D^+$ is a Morita-conservative expansion over the admissible site (the induced functor on structural model categories is an equivalence compatible with the site topology and target fiber), then:*

$$F \text{ has effective descent in } \mathcal{L}_D \iff e_*F \text{ has effective descent in } \mathcal{L}_D^+.$$

Proof sketch. Conservative presentations induce equivalent descent categories only when the equivalence preserves the site topology: covering families, pullback compatibility, and the target fiber are transported together. Equivalences preserve existence and uniqueness of global objects. Effective descent is exactly existence of a global object realizing the local datum. No identification-failure premise is used. $\square$

Corollary. *Conservative expansion cannot overturn $Z = 1$. If the target fiber has non-effective descent, no Morita-equivalent re-presentation of the structural language changes this. Any future argument that a richer structural language supplies the sufficiency bridge must add a non-conservative*

admissible operation. Such an operation changes Y , and Ω must declare it. Treating it as a language-only clarification will not do.

5.3 The descent criterion

Proposition (Finite deterministic factorization). *Let H and $\mathcal{R}_Y$ be finite sets, let $E_Y : H \rightarrow \mathcal{R}_Y$ be a deterministic observation map, and let $\Theta : H \rightarrow T$ be a target with decidable equality. An identification procedure using only admissible records exists if and only if there is a map $\hat{\Theta} : \mathcal{R}_Y \rightarrow T$ such that*

$$\Theta = \hat{\Theta} \circ E_Y.$$

Equivalently, Θ is constant on every fiber of E_Y .

Proof. If such $\hat{\Theta}$ exists, the procedure observes $r = E_Y(h)$ and returns $\hat{\Theta}(r)$. If an identification procedure using only records exists, its output is a function of r alone. Two hidden states in the same fiber of E_Y therefore receive the same output, so correctness requires Θ to be constant on that fiber. Define $\hat{\Theta}(r)$ to be that common value for each realized record. Unrealized records can be assigned arbitrarily. This is the finite deterministic form of effective descent: local record data assemble a unique global target value exactly when the target factors through the record object. $\square$

Proposition (Exact finite stochastic factorization). *Let H and $\mathcal{R}_Y$ be finite sets. Let $K : H \rightarrow \Delta(\mathcal{R}_Y)$ assign each hidden state an exact observation distribution over records, and let $\Theta : H \rightarrow T$ be a target with decidable equality. An exact identification procedure whose information is exhausted by the stochastic observation site exists if and only if Θ factors through the kernel map:*

$$\Theta = \hat{\Theta} \circ K$$

for some map $\hat{\Theta}$ defined on the image of K . Equivalently, whenever $K(h) = K(h')$, $\Theta(h) = \Theta(h')$.

Proof. If Θ factors through K , the procedure reads the exact kernel $K(h)$ and returns $\hat{\Theta}(K(h))$. Conversely, any exact procedure using only the stochastic site must assign the same answer to hidden states with the same observation distribution. Correctness therefore requires Θ to be constant on each fiber of K . Define $\hat{\Theta}$ on each realized kernel by that common value. $\square$

This proposition concerns exact distributional information. It does not say that one finite sample identifies the target. If two kernels overlap, one-sample classification can have nonzero Bayes error even when the exact kernels are distinct. A sharper obstruction is at work here: if the kernels are identical and the targets differ, no amount of sampling from that site identifies the target.

Grothendieck's theory [Grothendieck et al., 1971] establishes that effective descent over a site is exactly the condition under which coherent local data determine a global object. The propositions above prove the reduction for finite deterministic systems and exact finite stochastic systems. Every identification procedure we know of (maximum-likelihood estimation, instrumental variables, Pearl's do-calculus [Pearl, 2009], Bayesian updating) takes the same form: given local measurement records that cover the state space at some observational resolution, recover the target parameter. That recovery has the structure of effective descent of the target fiber over the measurement site. So when the target fiber has non-effective descent, no reconstruction procedure of this form should succeed, because the local data carry too little information to assemble a global answer. The reduction is argued here and proved for the formal components named in §13. A full proof that every admissible

identification procedure in every substrate class factors through effective descent lies beyond this article.

Fix a site (C_Y, J) , where J is a Grothendieck topology on the category of admissible measurement contexts. Non-effective descent of the target fiber F over this site says that no compatible family of local sections uniquely assembles into a global section. One invariant, non-effective descent of the target fiber, runs across every substrate, even though the computational tool that detects it shifts with the topology. Classical finite systems call for Čech H^1 , quantum POVM systems for the Abramsky–Brandenburger empirical-model method [Abramsky and Brandenburger, 2011], and coalgebraic systems for bisimulation-invariant descent. A substrate-specific table below lists these implementations. The claim that each substrate instantiates the same descent criterion under its natural topology is a scoped mathematical program, one this article does not complete.

Theorem 1 (General descent-obstruction criterion, conditional on the descent-factorization obligation). *Let (H, Y, E_Y, Θ) be an experimentable theory with declared revision envelope Ω . Suppose the descent-factorization obligation holds for the substrate: admissible identification procedures factor through effective descent of the target fiber over the observation site. This obligation is proved for the finite deterministic and exact finite stochastic cases above. It remains open in general (§13) and is assumed established nowhere else in this paper.*

Then Θ is necessarily non-identified from admissible observations if and only if $Z = 1$.

Proof sketch under the descent-factorization obligation. If $Z = 1$, then every admissible revision $r \in \text{Rev}_\Omega(Y)$ gives non-effective descent. No reconstruction of Θ from observable records exists for any admissible protocol or structural revision within Ω , since descent is the most general reconstruction principle and the Proposition (Conservative Invariance) rules out conservative expansion as a route around the obstruction. If $Z = 0$, some revision r gives effective descent, so the target fiber can be reconstructed from local data via the descent morphism. $\square$

The non-applicability boundary can be stated formally. A substrate is in-domain (the theorem applies) if and only if all four of the following are defined: (1) an admissible context site C_Y , (2) a target-fiber object $F \rightarrow H$, (3) descent semantics for that substrate category, and (4) a declared revision envelope $\text{Rev}_\Omega(Y)$. If any term is absent, the correct output is non-applicability (neither $Z = 0$ nor $Z = 1$) and the governance default is fail-closed.

5.4 The sufficiency boundary

Theorem 1 says when P1–P4 systems lie in the non-identification class under the descent-factorization obligation. Whether every P1–P4 system is conscious, the sufficiency direction, we settle here as a formal boundary result.

Conservative expansions supply no sufficiency. Suppose a future structural language $\mathcal{L}_D^+$ is conservative over $\mathcal{L}_D$ (Morita-equivalent, by the Proposition above), so that it draws no new structural distinctions between models:

$$\text{Mod}(\mathcal{L}_D^+) \simeq \text{Mod}(\mathcal{L}_D) \implies \Delta(\text{distinguishable structural cases}) = 0.$$

So a conservative expansion can turn $\text{P1–P4} \not\Rightarrow \Theta$ into $\text{P1–P4} \Rightarrow \Theta$ only if the implication was already derivable from the old structure, which the current framework nowhere licenses. Conservative re-presentations of the language leave the sufficiency bridge unbuilt.

Non-conservative expansions change the domain. Suppose $\mathcal{L}_D^+$ adds a primitive that directly tracks Θ . That clarifies no language, since it adds a new admissible probe and changes Y . A sufficiency bridge derived from $\mathcal{L}_D^+$ would then follow from the new Y , with the old structural framework doing none of the work. Any such expansion must be declared in Ω and evaluated as a non-conservative revision.

So the negative sufficiency result, that P1–P4 systems are not thereby shown to be conscious, owes nothing to an impoverished structural language. It holds under every conservative expansion, and only a declared non-conservative revision that changes what is admissible can defeat it. Within any observable-relative framework, the bridge from the non-identification class to consciousness attribution calls for an exogenous commitment that lies outside the framework. Other axiomatizations may add further commitments. The scope here is only what observable-relative formal frameworks can determine before any such commitment is made.

5.5 Comparison of approaches

Table 1 summarizes how the kernel-pair lemma, cohomological diagnostics, and the descent theorem cover the representative substrate classes used in the paper.

Case	Lemma 1	$\check{H}^1$	Thm. 1
Two states look identical, Θ differs	catches	misses	catches
Contextuality triangle (classical)	misses	catches	catches
Quantum POVM contextuality	n/a	catches	catches
Non-abelian symmetry system	partial	partial	catches
Identifiable control	silent	silent	silent

Table 1: Coverage across substrate classes. Lemma 1 is the H^0 special case of Theorem 1. $\check{H}^1$ is a sufficient diagnostic under classical good-cover conditions only.

In practice the right mathematical tool depends on the substrate:

Substrate	Use this obstruction tool
Classical finite systems	Čech H^1 ; CSP global-section check
Quantum (POVM) systems	Abramsky–Brandenburger empirical model [Abramsky and Brandenburger, 2011]
Non-abelian systems	Non-abelian H^1 , torsors, groupoid descent
Measurable/probabilistic	Invariant σ -algebras, measurable descent
Coalgebraic / process systems	Bisimulation-invariant descent

Table 2: Substrate-specific tools for computing the descent-obstruction flag.

Appendix A and the experiments directory make the checks below executable.

- *Contextuality triangle*: δ^0 rank over $\mathbb{F}_2 = 2$; $\dim \check{H}^1 = 1$; zero global sections; $D = 1$.

- *Kernel-pair case*: $\check{H}^1 = 0$; $D = 1$; H^1 -only recall on two-case suite = $1/2$.
- *Identifiable control*: $D = 0$.
- *Finite stochastic kernel pair*: identical observation kernels with different targets fail exact factorization; distinct kernels pass exact factorization while retaining finite-sample error.
- *Boolean recurrent architecture*: report-only traces leave a latent target unidentified across all checked input sequences; a direct latent reader removes the obstruction by changing the site.

In the consciousness applications below, the simpler branches do most of the work. IIT fails already at the witness stage, while GWT and higher-order theories raise kernel-pair worries whenever access or metacognitive evidence serves a phenomenal target. The descent generalization comes into its own on contextual, quantum-adjacent, and non-abelian substrates, where those simpler diagnostics no longer cover the space of identification failures.

6 The governance corollary

When $Z = 1$ for a system, no experiment (no certificate, audit, or disclosure protocol) can establish that Θ is identified. An epistemic claim and a normative premise follow, and they stay apart.

The epistemic claim flows from Theorem 1 alone. A governance protocol that issues a low-concern verdict on consciousness for such a system issues a verdict it cannot substantiate, because the theorem shows that no admissible experiment can supply the required identification evidence.

The normative premise: a low-concern verdict requires epistemic support, and where no support is possible the appropriate governance default is to restrict. The premise earns its place from the asymmetry of error costs under consciousness uncertainty, which warrants precautionary closure. Being conditional, it dominates only when a mistaken low-concern verdict would be severe, hard to reverse, and borne by a possible subject, while the cost of withholding the verdict stays bounded by delay, further disclosure, narrower deployment, or a graduated precautionary regime. Where those conditions fail, the mathematics still delivers the epistemic negative (the low-concern verdict lacks identification evidence) while leaving the practical policy response open. Given the premise, restriction follows as a theorem. Without it, the mathematics still shows that low-concern verdicts are epistemically unwarranted, yet says nothing about the governance response. The criterion's normative force stays confined to fail-closed audit and governance settings. It makes no claim that non-identification is inadmissible for science in general.

Corollary 1 (Fail-closed verdict rule). *If $Z = f(X, Y) = 1$, then for every admissible $Y' \in \text{Adm}(Y)$:*

$$\Theta \notin \text{Desc}(Y') \implies \text{low-concern verdict on } \Theta \text{ is not admissible.}$$

Proof. By $Z = 1$ every Y' gives non-effective descent. A low-concern verdict requires establishing identifiability under some Y' . Non-effective descent entails no such demonstration exists. $\square$

A verdict can come out three ways.

Situation	Correct verdict
Experimentability witness fails	Non-evaluable $\rightarrow$ no low-concern verdict
Witness holds and $Z = 1$	Identification impossible $\rightarrow$ verdict formally forbidden
Witness holds and $Z = 0$	Possible $\rightarrow$ requires adversarial certificate + replication

Both the “witness fails” and “ $Z = 1$ ” branches are fail-closed, so a governance protocol open to low-concern verdicts in either branch contradicts Corollary 1.

Most governance frameworks for AI consciousness are precautionary, advising caution because the evidence is incomplete. The corollary isolates a narrower case. When $Z = 1$ relative to a declared Ω , the declared evidence base simply cannot identify the target. On the normative premise adopted above, a low-concern verdict then lacks the support it requires.

In its audit form the corollary is direct. A low-concern verdict must state the site, target, revision envelope, classifier, challenge record, revocation rule, and provenance record it relies on. Should any required field go missing, the verdict is not evaluable under this criterion. As a downstream policy matter beyond this paper, the split between the access/report target and the phenomenal target supports a deployment license tied to the access/report verdict alone and barred from carrying that verdict over to phenomenal status.

6.1 The disclosure requirement

A protocol can apply the governance rule only once the system’s descent structure is on the table. A low-concern claim that omits the following six elements is not evaluable. It receives the same fail-closed verdict as $Z = 1$.

Any organization asserting a low-concern verdict for Θ must publish:

1. The context category/site C_Y : a complete specification of admissible measurements and interventions.
2. The target fiber object $F \rightarrow H$: a structural specification of Θ as a fiber over the hidden-state space.
3. The obstruction datum X : the descent data (transition maps, holonomy, kernel-pair constraints, contextual compatibility) that enables a third-party auditor to evaluate whether the system falls in the P1–P4 class.
4. The admissible revision envelope Ω : the declared physical, computational, and operational constraints on structural revision, with explicit classification of each proposed language expansion as conservative (Morita-equivalent) or non-conservative (genuinely new admissible probe or operation).
5. A revision-stable descent test result: a demonstration that $Z = \min_{r \in \text{Rev}_\Omega(Y)} D(r_* X) = 0$, with a certificate replicable by a third party under the declared Ω .
6. A non-applicability statement or confirmation: an explicit declaration that all four in-domain conditions (§5.3) are satisfied, or identification of which condition fails and why non-applicability, not $Z = 0$, is the correct output.

Any missing element forces fail-closed. Without Ω , Z is not evaluable under the revision-stable criterion. Undeclared non-conservative expansions then let a claimant revise the admissible site after the fact to dodge a $Z = 1$ verdict.

Each element guards against a specific governance failure mode. Undeclared site data lets a claimant gerrymander the admissible evidence base. Undeclared holonomy data hides whether the system falls in the P1–P4 class. A missing Ω opens the door to strategic omission of the non-conservative

revisions that would resolve the obstruction. And a missing revision-stable certificate licenses unilateral identifiability claims that no one can audit as the structure changes.

6.2 The Ω -audit protocol

Naming Ω as an input, as the disclosure requirement does, leaves it open to gaming. A claimant who wants a low-concern verdict can shape the envelope before the descent test runs, and that choice can decide whether Z is evaluable at all. An Ω declaration therefore needs its own audit protocol.

The protocol is organized around six failure modes. They need not exhaust the space. They form a finite audit framework against which a later completeness theorem could be tested.

Under-specification. A claimant may declare Ω vaguely, for example by saying that admissible revisions are those consistent with current physics or current practice, which leaves later challengers to be accepted or excluded after the fact. To remedy this, enumerate finitely. The declaration must list admissible revision classes (new observable types, intervention types, model-access regimes, target-relevant operations, and structural vocabulary changes), each carrying an inclusion or exclusion rationale. Anything outside the enumeration is inadmissible for the current certificate, though the enumeration itself stays open to public challenge.

Strategic exclusion. A claimant may exclude a revision class that would resolve the obstruction by calling it physically impossible, operationally infeasible, or out of scope. An exclusion is legitimate only when the reason is typed and auditable. A finite list of reason types is admissible: physical impossibility, computational intractability, regulatory prohibition, or cost above a declared threshold, where the threshold itself is part of the claim. Let an excluded class lack one of these reasons, or let the reason fail under audit, and the low-concern verdict is revoked or reduced to non-applicability. Each exclusion rationale is negative, recording why a class stays out. The claimant must also give its positive counterpart, the minimal set of non-conservative probes that would defeat the obstruction and return $Z = 0$ if they were admissible and run. Fixing that set before certification blocks a later denial that a surfaced probe was ever relevant. This does not certify completeness, since the falsifying set need not be finitely nameable in advance, but it closes one route to revising the claim after a challenge.

Misclassification. A claimant may add a probe that changes the admissible site while calling it a conservative language expansion. The audit rule is simple: any expansion claimed as conservative must supply a witness that the new presentation preserves the target fiber, covering families, and context structure. In mathematical cases this is a Morita-equivalence witness, and in applied cases the corresponding operational equivalence argument. With no witness, the expansion counts as non-conservative and triggers a fresh Z evaluation on the changed site. For finite structural fragments, where the equivalence is decidable, the witness must be machine-checkable, as the finite-alias case already is in Lean (§13), so a third party can recompute the equivalence and need not take the claimant’s reading on trust. The equivalence is undecidable in general, and there the witness reverts to a refereed proof, with the fail-closed default holding when none is supplied. That referee pool is part of the apparatus the provenance certificate covers. Fail-closed also blunts an attack by attrition: a claimant who floods the review with proofs too large to check leaves the referees unable to accept the witness, so the default holds and the expansion counts as non-conservative. Only an affirmative, captured certification can pass a site-changing probe off as conservative. Referee exhaustion cannot.

Post-hoc revision. A claimant may alter Ω after seeing data or after receiving a challenge. Here the remedy is preregistration discipline. The envelope must be declared before certification, timestamped,

and hashed. Any later change starts a new certificate. The earlier verdict inherits no evidence from the revised site until the new evaluation runs.

Residual discovery failure. A claimant may satisfy the listed declaration rules while a material non-conservative probe class is never surfaced. Seeded omission tests do not solve this. They can show that a red team detects known planted defects under benchmark conditions, but they cannot estimate how likely an adversarially chosen future probe class is to remain outside the audit community’s search space. So seeded omissions function as a skill gate, leaving residual closure uncertified. For operational use, write ε_{esc} for the residual adversarial escape load and ι_{audit} for verified audit integrity: independent selection, escrow funding, access parity, public challenge logs, conflict checks, and whistleblower channels. The admissibility score has the form

$$\sigma_{\text{adm}} = \iota_{\text{audit}}(1 - \varepsilon_{\text{esc}}).$$

Whenever a required dossier field is absent, access parity fails, red-team independence is compromised, or ε_{esc} cannot be bounded under the declared adversary model, the output is non-applicability. A seeded-omission success rate is no admissible bound for ε_{esc} .

One concrete instantiation of the declared adversary model is an escrow stake. A claimant seeking a low-concern verdict places capital or compute in escrow that funds independent red teams, who earn the stake by exhibiting an admissible probe that flips the verdict within the declared rules, which then revokes the certificate. This sharpens discovery by giving the search real resources and a real incentive. It does not certify closure. A quiet escrow period, with no probe found, shows only that none was found under that search and is no bound on ε_{esc} , so the residual-closure requirement above still holds.

Apparatus provenance failure. A claimant may shape the people or infrastructure that generate the challenge record before the public record is fixed. The visible dossier can still display formal independence, escrow funding, access parity, public logs, and a timestamped envelope, even as the hidden dependence sits upstream of all those records. The channels at issue include funding, employment prospects, advisory ties, compute or API access, benchmark infrastructure, data access, publication access, shared investors, and informal reciprocal arrangements. Write this residual as X_{prov} . A low-concern verdict therefore calls for a provenance certificate covering auditors, red teams, classifiers, panel members, infrastructure providers, and benchmark or data providers. The certificate must aggregate claimant-originated benefits by source class. Per-edge materiality thresholds may cap how much detail is published, yet they shelter no one from aggregation. Where source-class provenance cannot be certified, the provenance gate fails and the output is non-applicability.

All of this shifts the burden of a low-concern verdict onto the claimant, who must publish the envelope before the descent test, defend each excluded revision class, supply equivalence witnesses for conservative expansions, certify the provenance of the challenge apparatus, preserve challenger independence, and accept revocation when the envelope was too vague or moved after inspection. A third-party auditor need not settle consciousness. The auditor asks only whether a fixed and contestable revision envelope supports the claimed low-concern verdict.

Those six failure modes are the main attacks this proposed protocol blocks. A later reduction theorem would show that every strategy for gaming an Ω declaration either leaves the envelope under-specified, strategically excludes a relevant revision class, misclassifies a site-changing operation as conservative, revises the envelope after evidence is observed, exploits a failure of the discovery apparatus, or contaminates the provenance of the apparatus that generates the public record. No such reduction theorem is claimed here. If such a reduction holds, the audit protocol is complete

relative to the declared site and target fiber. If it fails, the counterexample identifies another failure mode and the protocol must be expanded. Hardest of all is the candidate counterexample of a future probe class that cannot be named at declaration time. A completeness proof must show that this reduces to one of the public-record failure modes, or else treat it as residual discovery failure. Operationally, this means that a low-concern dossier must not infer residual closure from planted omissions. It must either bound residual escape under a declared adversary model and a public challenge process, or fail closed. A second counterexample is provenance atomization, where a claimant splits influence across sub-threshold proxy paths so that no single edge looks material even as the aggregate claimant-originated exposure is material. The protocol blocks this only when the provenance certificate aggregates by source class. A five-mode reduction that files all such cases under residual discovery sweeps too widely, classifying by failed outcome where it should classify by a pre-specified mechanism. A reduction theorem over these six public objects is the natural formal target for follow-on work: declaration content, exclusion boundary, conservative classifier, certification timing, discovery apparatus, and apparatus provenance.

7 A worked application: a minimal broadcast-report architecture

Consider a small cognitive architecture. It is not a model of biological consciousness. It shows how to specify the site, target fiber, holonomy, and P4 condition in a case where one can check the obstruction by inspection.

The architecture has a perceptual store P , a workspace W , and a report controller R . The hidden state $h \in H$ contains a latent episode token and three local encodings of that token, one in each module. The admissible observation site C_Y has three contexts:

$$U_{PW}, \quad U_{WR}, \quad U_{RP}.$$

Each context exposes pairwise agreement records between two modules. No admissible probe reads all three encodings at once. This restriction is an ordinary instrumentation limit. The available protocols can test pairwise access, workspace availability, and report alignment, yet no single record identifies the latent episode token across all modules at once.

The target Θ is global episode identity: whether the three local encodings are assignments of one and the same latent episode. The target fiber $F \rightarrow H$ carries a local binary label for the episode token in each context. On overlaps, local labels are related by transition maps. Take two transition maps to preserve the label and one to flip it:

$$g_{PW,WR} = +1, \quad g_{WR,RP} = +1, \quad g_{RP,PW} = -1.$$

The product around the cycle is -1 . Every pairwise comparison is locally satisfiable, yet no consistent global label survives across all three contexts. The target fiber therefore has non-effective descent over C_Y , and Online Resource 1 runs the finite check.

With this in hand the P1–P4 check becomes explicit. P1 holds because each pairwise context supports a local assignment of the episode label. P2 holds because overlap assignments are related by transition maps, and the cycle product prevents global compatibility. P3 holds because the induced context-groupoid action has no global fixed section. Equivalently, transporting a local label around the cycle returns its opposite. P4 holds relative to the following declared revision envelope: admissible revisions may add time resolution, duplicate pairwise probes, replace labels by conservative aliases, or add pairwise behavioral readouts, but may not add a direct all-context

token reader. Each admissible revision commutes with the transition maps and preserves the cycle product -1 . Thus no revision inside Ω dissolves the obstruction.

What would count as a genuine defeat of the obstruction also shows up here. A probe that reads the latent episode token jointly across P , W , and R changes the site. Such a probe is a non-conservative revision, and Ω must declare it. Where it is physically and operationally admissible, the case moves from $Z = 1$ to $Z = 0$, or to a new site requiring a fresh computation. Otherwise the low-concern verdict stays unsupported.

This example generalizes. P1–P4 are no mere labels pinned on after the fact. They are a checklist: context site, target fiber, transition maps, holonomy, and revision envelope. They also pry access apart from phenomenal attribution. The architecture can show pairwise broadcast and report alignment while still failing to identify the global target. That is the form of the test the later theory sections apply to IIT, GWT, and HOT at a higher level of abstraction.

A second executable check trades the pairwise-context cycle for a small Boolean recurrent architecture. The system has an input bit, an observable report bit, a memory bit with update rule $m_{t+1} = x_t$, and a latent bit with update rule $a_{t+1} = a_t$. Across every input sequence up to the checked horizon, the report-only site sees identical traces for two states that differ only in the latent bit. A direct latent reader resolves the target, but only by changing the site. Online Resource 1 supplies the check as `boolean_recurrent_obstruction_check.py`.

8 A worked disclosure dossier: Llama 3.1 8B-Instruct

This dossier is a positive control: it illustrates how the framework applies to a public system without claiming LLM consciousness or its absence. It accepts ordinary access/report evidence for a narrow target while declining to carry that evidence over to phenomenal consciousness, and it applies the disclosure requirement to a public system in place of a toy architecture. The system is Meta Llama 3.1 8B-Instruct, an open-weight, text-only, instruction-tuned transformer released in July 2024 [Grattafiori et al., 2024, Meta, 2024]. The model card describes it as an autoregressive language model using an optimized transformer architecture, with supervised fine-tuning and RLHF for helpfulness and safety alignment. For the 8B model, the public record states a 128k context length, grouped-query attention, multilingual text input, text and code output, pretraining on more than 15 trillion tokens, and a December 2023 pretraining data cutoff.

The dossier is deliberately narrow. It concerns one self-hosted inference instance of the released 8B instruction model, with no external tools, memory, autonomous action loops, fine-tuning, or agent scaffolding. Any of those additions would change the site and call for a fresh dossier.

Mechanistic interpretability already offers many ways to test whether representations in a transformer track prompt content and shape later output. The dossier’s contribution is the audit distinction. Evidence that the model uses a context item is evidence for an access/report target. On its own that evidence leaves the phenomenal target untouched, and the two targets stay apart when a verdict issues.

1. Observation site. The admissible site contains: public model-card information; released model weights and architecture; token-level input-output behavior; prompt interventions; logit traces; activation probes; ablations of attention heads, MLP blocks, layers, residual streams, and safety wrappers; and reproducible self-hosted inference logs. It excludes non-public training traces, vendor-internal human-feedback records, private safety-classifier decisions, and unlogged deployment-layer

state. It also excludes biological measurements, because the system is not biological.

2. *Target fiber.* Access and phenomenal status are separate targets. For access or report, the target fiber holds operational labels such as “information is available for token generation,” “the model reports a state,” or “the model can use a context item in later output,” and ordinary behavioral and mechanistic probes can test these. For phenomenal consciousness or moral patienthood, the target fiber holds a phenomenal-status label for an inference state, with the public site supplying no decidable equality test for it.

3. *Revision envelope.* The declared Ω permits longer prompt batteries, adversarial prompts, activation patching, sparse-feature probes, causal ablations, alternative decoding parameters, quantization checks, and removal or replacement of deployment guardrails. It also permits comparison across self-hosted replicas of the same released weights. It forbids adding persistent memory, tool use, online learning, multi-agent scaffolding, biological sensors, or private training data while still calling the result the same site. Such changes are non-conservative and require a new Z evaluation.

4. *Classifier.* For access/report targets, the classifier is behavioral and mechanistic: does a probed representation causally affect later token generation or report behavior under held-out prompts? For phenomenal consciousness, the public record supplies no accepted classifier. Self-reports about consciousness are themselves outputs under the same observation site, never labels for the hidden target.

5. *Challenge record.* A sufficient public challenge record would include prompt suites, activation probe protocols, ablation results, negative controls, model/version hashes, sampling parameters, deployment-wrapper descriptions, and failed attempts to find a non-conservative probe that changes the verdict. The current public record holds useful model-card and benchmark information, but as a consciousness-attribution challenge log it falls short.

6. *Revocation rule.* A verdict must be recomputed once the site changes, as it does under tool integration, persistent memory, online adaptation, fine-tuning, changed safety wrappers, multimodal inputs, autonomous action loops, or access to private training traces. It changes too if a future theory supplies a decidable phenomenal classifier over this class of systems.

7. *Provenance record.* A low-concern certificate would oblige the audit to name who selected the probe classes, who ran the red-team process, who maintained the classifier, which infrastructure was used, and whether those actors or facilities carried claimant-originated dependencies. Here the provenance record stays intentionally limited: the author selected and ran the probe packet, the GPU infrastructure was rented, the model came from an ungated public mirror after the official gated repository proved inaccessible, and no independent red-team or source-class provenance certificate was produced. This suffices for a worked access/report illustration. For a low-concern phenomenal-consciousness certificate it does not.

Verdict. For the access/report task in this dossier, the result is evaluable and yields $Z = 0$: one can probe, ablate, and compare the public model across replicates for that target. For phenomenal consciousness or moral patienthood, the dossier is non-applicable. The ground for that is the public site’s lack of a target fiber with decidable equality, a phenomenal classifier, a public consciousness challenge log, and the provenance certificate a low-concern verdict requires. It is no demonstration that Llama 3.1 8B-Instruct is non-conscious. So the framework blocks a certified low-concern claim for the phenomenal target under this dossier while letting ordinary scientific claims about access, report, and mechanism through.

The dossier infers no consciousness from opacity. It separates targets, treats the system as experimen-

tally tractable for access and report, and refuses only the stronger phenomenal verdict, on the ground that the dossier does not contain the objects such a verdict requires. The executable access/report probe packet is `llama31_activation_probe.py` in Online Resource 1. It freezes the prompt schema and declared site in a manifest, then on a GPU collects selected hidden states, reports leave-one-out probe accuracy by layer, and runs a targeted residual-direction intervention. This packet bears on access/report evaluability and nothing more. On 2026-05-06 it ran on an A10 GPU against the ungated `NousResearch/Meta-Llama-3.1-8B-Instruct` mirror, using 24 controlled key-value prompts. Token-level answer accuracy was 24/24, with a mean correct-label logit margin of 6.107 against a no-mapping control margin of roughly zero (-0.026). Leave-one-out centroid probes on selected hidden states reached 1.00 accuracy at layers 16, 24, and 31, and 0.75 at layer 8, and a targeted residual-direction intervention reduced the correct-label margin at layers 16, 24, and 31 with mean deltas -0.732 , -1.669 , and -1.107 . These results support the narrow access/report claim under the declared site while supplying no phenomenal-consciousness classifier. Online Resource 1 holds the saved packet as `llama31_probe_packet_nous/summary.json`.

The experiment functions as a positive control. Linear probes and residual interventions are familiar tools for showing that transformer representations track prompt content and shape later output, so the experiment adds no new mechanistic-interpretability result. Its point lies in the audit discipline. The framework accepts the ordinary evidence for the access/report target, assigns it the expected evaluable verdict, and then declines to transfer that verdict to the phenomenal target until someone supplies the missing target fiber, classifier, challenge record, revision envelope, and provenance certificate. Had the access/report control failed, the dossier would have told us nothing even about the narrow target. Because it succeeds, the non-applicability verdict for the phenomenal target cannot be charged to a blanket skepticism about empirical AI systems.

9 Implications for IIT, GWT, and higher-order theories

These three cases do different jobs. IIT is the hardest case for the experimentability witness, since its central quantity is defined at a level that current measurement cannot reach. GWT and higher-order theories sit more comfortably as theories of access and metacognition, and their difficulty surfaces only when those narrower targets are pressed into service as evidence for phenomenal consciousness.

9.1 Integrated information theory

IIT identifies consciousness with integrated information Φ above a threshold [Tononi et al., 2016]. The live question is what one would have to measure to make a full Φ -based identification claim auditable, not whether IIT is true. Computing Φ calls for the cause-effect repertoire of every possible subsystem, across all spatial and temporal scales, which for a system of N binary neurons means 2^N perturbation experiments. No current neuroimaging or electrophysiology protocol comes close, so under any known experimental protocol the observation record $\mathcal{R}_Y$ lacks the information needed to compute Φ with decidable equality.

For the purpose of full Φ -based identification, IIT’s target Θ falls in the prior class, because the observation protocol required to test the target does not exist. So it is not yet in the “ $Z = 1$, identification impossible” class. Specifically, full cause-effect repertoires lie outside $\text{Adm}(Y)$ under any currently realizable experimental protocol.

One distinction needs marking. IIT’s *predictions* are falsifiable, and the 2025 adversarial study challenged one salient preregistered prediction [Cogitate Consortium et al., 2025] (Cogitate consortium,

$n = 256$, multimodal fMRI/MEG/iEEG): the expected sustained posterior synchronization did not appear. The same study also challenged GNWT, through the lack of ignition at stimulus offset and limited prefrontal representation of some conscious dimensions. Those results are scientifically relevant, yet they leave the identification question used here unsettled. The witness-failure verdict turns on a different question, whether one can compute the value of Φ from experimental data. Falsifying a theory-derived prediction needs only that the theory entail observable consequences, with no need to compute Φ . The identification problem arises only when no observation protocol can supply the quantity the theory defines as its target, Φ itself. Current approximation schemes such as Φ^* , MIB, and GeoMIP read best as proxies or restricted computations. The best implementations handle small systems far below brain scale, none of them cited here claiming decidable equality of $\hat{\Phi}$ with Φ for the full target.

The tractability objection. An IIT proponent may answer that Φ is computable in principle from complete cause-effect structure, the computation being exponential in N yet tractable in principle. The experimentability witness asks for something else, *decidable equality*: a procedure that definitively answers “does this measured quantity equal the theory-defined target Φ ?” That demand is distinct from tractability. To satisfy the witness one needs both (a) an observation protocol Y whose record $\mathcal{R}_Y$ contains the cause-effect repertoire, and (b) a decidable equality check between the measured quantity and the theoretical Φ . In practice the exponential-computation barrier blocks (a), and no current approximation scheme satisfies (b): each Φ^* variant is an acknowledged lower bound or proxy, far short of a decidable equality test against the theoretical target. So the witness-failure verdict rests on the absence of a decidable specification. [Hanson and Walker \[2019\]](#) independently established a related substrate-sensitivity problem for IIT, namely that Φ is not invariant under state-labeling isomorphisms, so functionally identical systems can receive different Φ values. The descent-obstruction analysis here supplies a substrate-neutral structural reason for that failure of invariance. IIT thus lacks the experimentability witness, meaning decidable equality on Θ , that a full Φ -based identification claim would need. A version of IIT specifying a practical approximation $\hat{\Phi}$ with a decidable equality test would clear this first hurdle, and whether $\hat{\Phi}$ actually tracks Φ would then become the next empirical question.

9.2 Global workspace theory

GWT identifies consciousness with global broadcast, information made available to multiple cognitive subsystems at once [[Baars, 1988](#), [Dehaene and Changeux, 2011](#)]. This target is far more tractable than Φ , since masking, report, attention, and neural-imaging paradigms can all test broadcast. For access consciousness in Block’s sense [[Block, 1995](#)], GWT can supply the experimentability witness. None of this lets current GNWT predictions off the hook: the Cogitate adversarial study challenged GNWT’s offset-ignition and prefrontal-content predictions. The narrower point holds anyway, that one can specify broadcast and access targets in an experimentable record type even where particular GNWT predictions fail.

Trouble begins once broadcast evidence is made to license claims about phenomenal consciousness. Two systems with identical global broadcast patterns may differ in phenomenal experience. Where the governance target is phenomenal consciousness, the kernel pair fires: (broadcast + phenomenal experience) against (broadcast + no phenomenal experience) produce identical records under any broadcast-based measurement. So GWT evidence can speak strongly to access while leaving the phenomenal target unsettled.

9.3 Higher-order theories

Higher-order theories identify consciousness with a representation of being in a mental state [Rosenthal, 2005, Carruthers, 2000]. Here again the first step is feasible: metacognitive reports, confidence judgments, and prefrontal-monitoring signals all fit a record type with decidable equality. The harder question is whether higher-order representation identifies phenomenality. Blindsight makes the problem vivid, with subjects showing information-guided behavior in the absence of ordinary visual experience. Should the kernel pair (higher-order representation + phenomenal experience) against (higher-order representation + no phenomenal experience) be empirically realizable, Lemma 1 applies. HOT therefore passes the witness for a metacognitive target, and passing it leaves phenomenal consciousness still unidentified.

10 The upload problem and evolutionary selection

Consider first the upload problem. Someone proposes a digital system D as conscious by copying the behavioral profile of a biological system B , so that $O(B) \approx O(D)$. Observable copying carries over the record $E_Y(h)$ while leaving behind the descent structure, the transition maps and holonomy of the target fiber. A verdict would require showing $(C_{Y_D}, F_D, \rho_D) \cong (C_Y, F, \rho)$. Behavioral matching alone does not reach that far, and absent the structural isomorphism the correct verdict for D is non-evaluable. Non-evaluability here does not deny the upload moral status. It blocks the verdict that would license disregarding the upload’s welfare, so under the precautionary premise the possible subject is protected, not cleared.

Consider next the evolutionary argument. Natural selection acts on observable outputs and reaches Θ only through them. Where Θ is descent-obstructed, selection on O does not entail selection on Θ . So the inference “consciousness evolved because it confers fitness advantage” becomes structurally invalid when $Z = 1$: selection may have favored hidden causal structures X that produce adaptive O , with the obstructed property Θ riding along as a byproduct.

The same machinery reads off a substrate-neutral identification claim. A digital system that instantiates an equivalent context category, transition maps, and holonomy structure, $(C_{Y_D}, F_D, \rho_D) \cong (C_Y, F, \rho)$, falls in the same identification class as a biological system with that structure. Descent equivalence across the declared sites is what matters. So the theory predicts that substrate-isomorphic systems are identification-equivalent once the site, target fiber, and revision envelope match.

A final caution: information flow is not holonomy. A system maximizing integrated information, global broadcast, or any other information-theoretic quantity satisfies P1, local contextual presentability in information terms, while it may well fail P3, nontrivial holonomy. Information-theoretic magnitude is measurable and identifiable. What generates the obstruction is structural holonomy over the admissible observation site, so large information flow sits comfortably with $Z = 0$.

11 Relation to the literature

We borrow the mathematics from elsewhere. Kernel-pair identification is standard in causal inference [Pearl, 2009], Čech cohomology for contextuality comes from Abramsky and Brandenburger [2011], and Grothendieck descent is sixty-year-old algebraic geometry [Grothendieck et al., 1971]. The contribution here brings this known machinery to bear on consciousness attribution.

Older and less tidy is the philosophical setting. Nagel [1974] and Levine [1983] made the epistemic difficulty of subjective experience hard to ignore, and Chalmers [1995] gave the difficulty its contemporary name. Dennett [1991] and Frankish [2016] press the other way, holding that what needs explaining may be the judgments, reports, and seeming of phenomenality, with no further hidden property posited. That dispute stays open here. The prior question is narrower: given a declared target, can one identify it from the admitted observations? Block’s access/phenomenal distinction [Block, 1995] marks exactly the place where many proposed AI tests quietly switch targets.

The approach is methodological and leaves the metaphysics aside. When the target is access, report, metacognitive monitoring, or functional integration, ordinary experimental and mechanistic tests may identify it. When the target is phenomenal consciousness in the sense at issue for Nagel [1974], Levine [1983], and Chalmers [1995], the claimant has to say how that target enters the observation site. An illusionist or behaviorist may instead deny that any further phenomenal target is needed, which legitimately changes the target. A replacement target is then evaluated in turn, and the discarded one drops out. None of this settles the dispute between realist and deflationary theories of consciousness. It is a discipline for moving between them: state the target, state the site, and show whether the target is identified.

The position is epistemic structuralism about scientific attribution, with no commitment on metaphysical structuralism about what exists. A realist may hold that there is a fact of the matter beyond the declared observations. That possibility stands, though the observation site does not license such a fact as a scientific verdict. An illusionist or behaviorist may hold that no further phenomenal residue is needed, which the analysis grants. Where the proposed target is report, judgment, seeming, or illusion-detection, that target is the one evaluated. The error to avoid is a careless use of Block’s distinction: access evidence can be valid evidence for access while leaving phenomenal consciousness unidentified. Which target one should care about is left open, with one exception, that a claimant must declare a change of target.

Philosophically, the question becomes one of identification: before treating evidence as decisive, a consciousness verdict must hold target, site, and admissible revision envelope fixed. Lemma 1 already gives a strong version of the point for classical indistinguishability cases, and Block’s access/phenomenal distinction explains why access evidence cannot simply migrate to the phenomenal target. The descent machinery carries the point to contextual and non-classical substrates where kernel-pair quotients or Čech H^1 are too narrow. Its fully general reduction remains a stated obligation, with the finite factorization cases and the conservative-invariance results establishing the part used here. From this follows a governance corollary carrying an explicit six-element disclosure requirement that includes the revision envelope Ω . From there we evaluate major consciousness frameworks against the experimentability precondition.

On formal frameworks for AI consciousness, Butlin et al. [2023] set the current methodological baseline by deriving computational indicator properties from GWT, IIT, HOT, and related theories and then assessing AI systems against them. Carefully empirical though it is, the approach presupposes that the indicator properties are experimentally accessible. Phillips and Tsuchiya [2024] restate all six IIT axioms as universal mapping properties in the sense of category theory, including the sheaf condition. The present framework extends theirs from definition to testability: when the sheaf condition fails (no global sections exist), that is exactly the descent-obstruction criterion we use as an identification-impossibility certificate. Kleiner and Tull [2021] give a categorical formulation of IIT aimed at defining consciousness, where our aim is to characterize when such definitions become scientifically evaluable. Gunji et al. [2026] give a recent categorical reconstruction of contextuality in

which orthomodular structure arises from classical Boolean contexts. That work is adjacent to our use of contextual obstruction, while our target is identification of a hidden property from admissible observations. Campero et al. [2025] provide the clearest taxonomy of impossibility-class arguments against AI consciousness, and our result contributes a new category, identification-level impossibility independent of substrate implementation. Phua [2025] use artificial agents and ablations to test functional predictions associated with GWT, IIT-adjacent markers, and HOT. Their dissociations support the need to keep access, integration, and metacognitive targets separate before making any phenomenal-consciousness verdict.

Closest of the adjacent prior work is Inoué [2026], who develops a sheaf-cohomological treatment of neural integration on brain manifolds, modeling local cognitive functions as sheaf sections and classifying integration failures as cohomological obstructions. The framework here differs from his in two respects. First, Grothendieck effective descent serves as the obstruction criterion in place of Čech H^1 . The kernel-pair example (Lemma 1) shows descent to be strictly more general: systems exist with $H^1 = 0$ but $D = 1$, so an H^1 -only diagnostic misses part of the identification-failure class. Second, the target is the identification problem, whether one can recover the property Θ from admissible measurements, where his is neural integration. The Proposition (Conservative Invariance) and the revision envelope Ω bear on the language-stability of the identification claim, a question Inoué leaves aside.

On governance under consciousness uncertainty, Long et al. [2024] and Butlin and Lappas [2025] argue that precautionary governance is warranted. The governance corollary sharpens their recommendation: for the hidden-variable substrate class, the uncertainty is identification-theoretic under the declared observation regime. On the normative premise adopted here, a low confidence score of “likely not conscious” falls short. Wolfson [2026] and Rost [2026] independently anticipate related protection and readiness concerns through non-mathematical reasoning. Birch [2024] develops a graduated precautionary framework for sentience attribution, and Corollary 1 gives a formal condition under which the evidential basis for low concern is absent. Where Zhou et al. [2025] defaults to a “presumption of no consciousness,” the present theorem rejects that default once descent obstruction holds and the governance target requires identification evidence. Under non-effectiveness, no negative verdict follows from the absence of admissible evidence.

12 Objections

The deepest objection concerns the phenomenal target itself. The descent test needs a target fiber with decidable equality, and for phenomenal consciousness no one has exhibited one, so a phenomenal $Z = 1$ would be either tautological or stipulated. The premise is granted. The conclusion does not follow. This paper claims no phenomenal $Z = 1$ for any real system. The $Z = 1$ results hold only where the target fiber is specifiable: the broadcast-report architecture under global episode identity, and access or report targets with decidable equality. Where the target is phenomenal consciousness, the public site supplies no such fiber, the descent test does not run, and the correct output is non-applicability, the fail-closed branch, not $Z = 1$. The claim for the phenomenal case is therefore diagnostic, not an impossibility theorem: consciousness science cannot at present state the phenomenal target in an auditable form, and until it can, a low-concern verdict has nothing to discharge. Non-identifiability is nowhere assumed in order to derive the verdict. What the framework diagnoses is non-specifiability, and that is what blocks the verdict.

A critic may object that Ω is a knob. Set it too narrow and the obstruction is shielded from serious future tests. Set it too broad and the result grows hard to evaluate. This is the framework’s

main practical pressure point. The answer is procedural: Z is always relative to a declared Ω , and the declaration forms part of the public claim, so a low-concern verdict carries an audit burden. The claimant must say which new observables, interventions, probes, and structural vocabularies are admissible, and why excluded revisions fall outside the physical or operational scope. An under-specified Ω yields neither $Z = 1$ nor $Z = 0$. It yields non-applicability and the same fail-closed governance result.

A behaviorist offers a different objection. If two systems are indistinguishable under every admissible observation, the behaviorist may deny that any remaining difference is scientifically meaningful. The objection has force at the epistemic level: when the target is not identified, science cannot license any target attribution, positive or negative, from the declared observations. The governance conclusion then adds a premise, that a low-concern verdict carries moral stakes and so requires identification evidence. The mathematics blocks the evidential claim. The normative premise fixes the default response.

A further objection presses on substrate. The framework is substrate-neutral only at the level of form, and substrate still does work through the declared inputs. For each substrate, the claimant must specify the admissible site Y , the target fiber $F \rightarrow H$, and the revision envelope Ω . A biological system, a digital system, and a quantum-contextual system may instantiate one descent pattern, but the claimant has to show the equivalence. Substrate enters through the site and the allowed revisions. Once those inputs are fixed, the theorem says what follows. It leaves the burden of fixing them where it lies, on the claimant.

13 Scope and limitations

Several things lie outside the reach of the argument. Which systems are conscious is one of them. The theorems characterize when science cannot answer that question. A verdict of $Z = 1$ never marks the absence of consciousness either, only the failure of admissible experiments to settle the matter. Nor does the §9 evaluation pronounce IIT, GWT, or HOT false: it is structural, and passes no truth judgment. The governance corollary, in turn, reaches some moral-status decisions and not every one. And a proof sketch is just that, never a proof.

The finite deterministic and exact finite stochastic factorization results are proved in §5.3. The broader reductions remain argued, not proved: that every admissible identification procedure across all substrate classes factors through descent, and that the descent criterion is computed identically across all five substrate classes in Table 2.

Eight claims here are mechanically verified in Lean 4 (Mathlib v4.30.0-rc2), each axiom-clean (no `sorry`, checked against an explicit axiom allowlist of `Classical.choice`, `Quot.sound`, and `propext`): the finite deterministic factorization biconditional ($E : H \rightarrow R$, image-restricted); the exact finite stochastic factorization biconditional ($K : H \rightarrow \text{PMF } R$, image-restricted, the harder of the two factorization results, closed as separate forward and backward lemmas); the finite kernel-pair non-factorization lemma; the finite-alias Conservative Invariance case; a categorical sheaf-transport statement under suitable equivalence; the Appendix A Case 4 Z -formula evaluations; the Čech-holonomy detection of §5.1 on an arbitrary finite nerve (a closed walk with nonzero holonomy admits no global section, for abelian coefficients, with the cyclic case as a special walk); and the revision-stability of that obstruction under cover refinement (subdividing each edge and redistributing its label leaves the holonomy invariant and nonzero, so no global section exists in any such refinement). This last result verifies revision-stability only under refinement of a fixed

cover. Stability under arbitrary change of measurement language stays argued, not proved. Online Resource 2 supplies the Lean files, which compile with only the standard axiom dependencies. They are verification receipts for named components, not a proof of the full descent-factorization reduction, which remains the main open reduction.

Operationalizing Ω is the principal open problem. A claimant must declare and audit the revision envelope Ω before Z is evaluable. For real physical and computational systems, specifying and publicly justifying Ω is open-ended. Practitioners may under-specify, strategically exclude plausible non-conservative revisions, or dispute which expansions count as conservative. This leaves the theoretical result untouched, since Z is well-defined relative to any declared Ω , but it lodges the evidential burden in Ω 's completeness and public auditability. The practical next step is therefore an Ω audit protocol: a declaration schema, an adversarial challenge process for missing revisions, and a revocation rule for when a low-concern verdict rests on an under-specified or strategically narrow envelope. Such a protocol must also refuse a common shortcut. Success on seeded omissions qualifies a challenger's competence while leaving unbounded the chance that adversarial probe classes remain unsurfaced. Residual escape must be bounded under a declared adversary model and public challenge process, on pain of non-applicability. The same holds for independence. A provenance certificate asks for more than the absence of a visible conflict: the dossier must certify the source-class provenance of auditors, red teams, classifiers, panel members, infrastructure providers, and benchmark or data providers, and it must aggregate claimant-originated benefits across proxy and sub-threshold paths. Lacking that protocol, the framework can still state non-applicability cleanly even while a real disclosure dossier stays too easy to game.

Treat internal adversarial review as a source of candidate objections. It does not certify the theorem. Evidence for the framework has to come from public proofs, public code, worked systems, or external audit records. Several such checks are now complete. §8 gives a public-system dossier for an open-weight transformer and a reproducible activation-probe packet for the access/report target; the finite stochastic script checks the exact kernel case; the Boolean recurrent script runs the obstruction calculation on a small transition system; and the prediction packet states the following falsifiable disclosure-packet prediction: for a fixed text-only, self-hosted, open-weight transformer with no external tools, memory, online learning, or autonomous action loop, access/report use of an input item should be evaluable from public weights, token-level behavior, logits, hidden states, activation probes, causal ablations, prompt interventions, and reproducible inference logs. Phenomenal consciousness or moral patienthood should remain non-applicable under that same site unless the claimant supplies a target fiber with decidable equality, a public classifier, a public challenge record, a declared revision envelope, and a provenance certificate for the challenge apparatus. The prediction fails if an auditor supplies those phenomenal-target objects without adding a non-conservative probe, or if the access/report packet cannot produce above-chance behavioral evidence for a system known to use the relevant input.

Several lines of further work remain. The public-system dossier should be extended to a deployed system with tool use, memory, and logged deployment state. The finite proof should be pushed from exact kernels to finite-sample statistical decision procedures and finite measurable systems before anyone claims the fully general version. The prediction packet should go to an external registry or independent auditor before it is used as external evidence. And the philosophy-of-mind engagement in §11 should continue by stating in each case exactly which target is at issue, whether access, report, integration, metacognition, or phenomenal attribution.

14 Conclusion

For some hidden-variable systems, consciousness attribution carries an identification problem that no further observation inside the declared site can remove. This happens when the target fiber has revision-stable non-effective descent. The systems in this class are characterized by four structural properties, P1–P4, that generate necessary non-identification for any substrate supplying the required site, target fiber, descent semantics, and revision envelope. Under the descent-factorization obligation, the revision-stable descent criterion (Theorem 1) states the impossibility, and the Proposition (Conservative Invariance) shows that conservative re-presentations of the structural language leave the obstruction in place. The framework’s boundary is also fixed: the bridge from P1–P4 to consciousness attribution calls for an exogenous commitment or a non-conservative revision of the admissible site. The governance corollary follows from all this. When $Z = 1$, a low-concern verdict lacks the identification evidence it requires, and an organization that issues such a verdict before satisfying the six-element disclosure requirement, the declared revision envelope Ω among them, has not supplied the evidence its verdict needs.

The framework becomes usable only once the claimant writes down the observation site, target fiber, transition data, and revision envelope for the system at hand. The worked application in §7 shows the shape of that specification on a small case. Without those objects, any consciousness verdict is premature.

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A Falsification suite

The code below confirms all the quantitative predictions and runs with no dependencies beyond Python’s standard library. The suite tests four claims: (1) the contextuality triangle has $D = 1$ and $\check{H}^1 = 1$; (2) the kernel-pair case has $D = 1$ and $\check{H}^1 = 0$, demonstrating $\check{H}^1$ -only recall = 0.5; (3) the identifiable control has $D = 0$; and (4) the revision-stability predictions, namely that a conservative language expansion leaves D unchanged and a non-conservative hidden-state-revealing revision removes the obstruction.

```
import itertools

def rank_f2(matrix):
    m = [row[:] for row in matrix]
    if not m:
        return 0
    rows, cols = len(m), len(m[0])
    rank = 0
    for col in range(cols):
        pivot = next(
            (r for r in range(rank, rows) if m[r][col] % 2 == 1), None)
        if pivot is None:
            continue
        m[rank], m[pivot] = m[pivot], m[rank]
        for r in range(rows):
            if r != rank and m[r][col] % 2 == 1:
                m[r] = [(m[r][c] ^ m[rank][c]) for c in range(cols)]
        rank += 1
    return rank

# Case 1: contextuality triangle
CONTEXTS = ["C12", "C23", "C13"]
CONTEXT_STATES = {
    "C12": ("h1", "h2"), "C23": ("h2", "h3"), "C13": ("h1", "h3")}

def overlapping_states(ca, cb):
    return set(CONTEXT_STATES[ca]) & set(CONTEXT_STATES[cb])

def local_value(ctx, state, section):
    return section[CONTEXT_STATES[ctx].index(state)]

def is_globally_consistent(sections):
    for ca, cb in itertools.combinations(CONTEXTS, 2):
        for state in overlapping_states(ca, cb):
            if local_value(ca, state, sections[ca]) != \
               local_value(cb, state, sections[cb]):
                return False
    return True

ODD_PARITY = {
    ctx: [(a,b) for a in (0,1) for b in (0,1) if (a^b)==1]
    for ctx in CONTEXTS}

def global_sections(local_family):
    return [
        {ctx: sec for ctx, sec in zip(CONTEXTS, combo)}
        for combo in itertools.product(
            *[local_family[c] for c in CONTEXTS])]
```

```

        if is_globally_consistent(
            {ctx: sec for ctx, sec in zip(CONTEXTS, combo)}}]

coboundary      = [[1,1,0],[0,1,1],[1,0,1]]
triangle_rank   = rank_f2(coboundary)           # 2
triangle_h1     = 3 - triangle_rank             # 1
triangle_gs     = len(global_sections(ODD_PARITY)) # 0
triangle_D      = 1 if triangle_gs == 0 else 0   # 1

assert triangle_rank == 2
assert triangle_h1   == 1
assert triangle_gs   == 0
assert triangle_D    == 1

# Case 2: kernel-pair ( $H^1 = 0$ ,  $D = 1$ )
obs      = {"a": "same", "b": "same"}
target   = {"a": 0, "b": 1}

def kernel_pair_D(obs, theta):
    for x, y in itertools.combinations(obs, 2):
        if obs[x] == obs[y] and theta[x] != theta[y]:
            return 1
    return 0

kernel_h1 = 0
kernel_D  = kernel_pair_D(obs, target)    # 1
assert kernel_h1 == 0
assert kernel_D  == 1

# Case 3: identifiable control
obs_ctrl  = {"a": "obs_a", "b": "obs_b"}
tgt_ctrl  = {"a": 0, "b": 1}
control_D = kernel_pair_D(obs_ctrl, tgt_ctrl) # 0
assert control_D == 0

# Detector comparison
total      = triangle_D + kernel_D
h1_detected = (1 if triangle_h1 > 0 else 0) + (1 if kernel_h1 > 0 else 0)
h1_recall   = h1_detected / total          # 0.5
descent_recall = total / total              # 1.0

assert h1_recall == 0.5
assert descent_recall == 1.0

```

Case 4 verifies a definition and runs no independent empirical test. By a category-theoretic argument, the Conservative Invariance proposition guarantees that adding a Morita-equivalent alias preserves the descent flag, while the non-conservative variant adds an observable that distinguishes the two hidden states and so, by the definition of the descent obstruction, must resolve it. Case 4 is included as a verification benchmark, confirming that the formal machinery behaves as the proposition predicts. It adds no evidence beyond what the proposition's category-theoretic content already establishes.

```

# Case 4: revision-stability benchmark
# Base: kernel-pair obstruction ( $D_{\text{base}} = 1$ ).
# Conservative revision: add a definitional alias of the existing observable.
# The Proposition (Conservative Invariance) predicts  $D$  unchanged.

```

```

# Non-conservative revision: add an observable that distinguishes the hidden
# states.
# This changes Y; the obstruction is resolved (D = 0).

def compute_D(obs_map, theta_map):
    for x, y in itertools.combinations(obs_map, 2):
        if obs_map[x] == obs_map[y] and theta_map[x] != theta_map[y]:
            return 1
    return 0

obs_base = {"a": "obs_X", "b": "obs_X"}
theta_base = {"a": 0, "b": 1}
D_base = compute_D(obs_base, theta_base)
assert D_base == 1, "Base kernel-pair must have obstruction"

obs_conservative = {"a": ("obs_X", "obs_X_alias"), "b": ("obs_X", "obs_X_alias")}
theta_conservative = theta_base
D_conservative = compute_D(obs_conservative, theta_conservative)
assert D_conservative == 1, "Conservative expansion must preserve obstruction"
assert D_conservative - D_base == 0, "Delta D for conservative revision must be 0"

obs_nonconservative = {"a": ("obs_X", "reveals_a"), "b": ("obs_X", "reveals_b")}
theta_nonconservative = theta_base
D_nonconservative = compute_D(obs_nonconservative, theta_nonconservative)
assert D_nonconservative == 0, "Non-conservative revision must remove obstruction"
assert D_nonconservative - D_base == -1, "Delta D for non-conservative revision
    must be -1"

conservative_overturns = 0
nonconservative_overturns = 1
actual_conservative_overturns = 1 - D_conservative
actual_nonconservative_overturns = 1 - D_nonconservative
assert actual_conservative_overturns == conservative_overturns
assert actual_nonconservative_overturns == nonconservative_overturns

n_generators = 10
m_hidden_states = 100
revision_subsets = 2 ** n_generators
hidden_pairs = m_hidden_states * (m_hidden_states - 1) // 2
total_checks = revision_subsets * hidden_pairs
assert revision_subsets == 1024
assert hidden_pairs == 4950
assert total_checks == 5_068_800

if __name__ == "__main__":
    print("All assertions passed.")
    print(f"Triangle: rank={triangle_rank}, H1={triangle_h1}, "
          f"global_sections={triangle_gs}, D={triangle_D}")
    print(f"Kernel-pair: H1={kernel_h1}, D={kernel_D}")
    print(f"H1-only recall={h1_recall} | Descent recall={descent_recall}")
    print(f"D_base={D_base}, D_conservative={D_conservative}, "
          f"D_nonconservative={D_nonconservative}")
    print(f"Conservative overturns={actual_conservative_overturns}, "
          f"Non-conservative overturns={actual_nonconservative_overturns}")
    print(f"Revision audit cost (n={n_generators}, m={m_hidden_states}): "
          f"{total_checks:,} pair-checks")

```
